

Acute alkalinity stress induces functional damage and alters immune metabolic pathways in the gill tissue of spotted scat (*Scatophagus argus*)

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ARTICLE INFO

Keywords:

Alkalinity stress
Scatophagus argus
Histology
Biochemical
Transcriptome

ABSTRACT

The spotted scat (*Scatophagus argus*), known for its robust osmoregulatory abilities as a euryhaline fish, represents a promising candidate for large-scale aquaculture in saline-alkaline waters. In this study, spotted scat were exposed to control conditions and a treatment group with alkalinity at 30 mmol/L for varying durations (0 h, 12 h, 24 h, 48 h, 72 h, and 96 h). Histopathological analysis, biochemical assays, and transcriptomic evaluation were employed to examine the impact of alkalinity stress on gill tissue. The results indicated that gill tissues exhibited varying degrees of damage when compared to the control group across different time points. Prolonged alkalinity stress resulted in an increased number of chlorine-secreting cells and caused the gill lamellae to curve, display inconsistent orientation, and swell at the base. Biochemical analyses revealed that acute alkalinity stress disrupted oxidative balance, leading to decreased activities of superoxide dismutase (SOD) and catalase (CAT), with a notable minimum observed at 96 h. In contrast, the activities of glutathione peroxidase (GSH-Px), acid phosphatase (ACP), alkaline phosphatase (ALP), and malondialdehyde (MDA) concentration gradually increased with extended alkaline exposure. Transcriptomic analysis of gill tissues from the control and 24 h acute alkalinity stress groups revealed significant changes in gene expression, identifying 513 differentially expressed genes (DEGs), with 292 up-regulated and 221 down-regulated. These DEGs were significantly enriched in pathways related to drug metabolism-cytochrome P450 and drug metabolism-other enzymes. Additionally, pathways involved in arachidonic acid metabolism, as well as arginine and proline metabolism, were found to play critical roles in adapting to alkalinity stress. This study contributes to a valuable insight into the molecular mechanisms underlying alkali resistance in spotted scat.

1. Introduction

In China, saline-alkaline land encompasses approximately 45.87 million hectares, with this area expanding at an alarming rate each year (Lin et al., 2013). As soil salinization intensifies, the available space for freshwater aquaculture is expected to progressively diminish (Ondrasek and Rengel, 2021). Moreover, the high salinity and alkalinity of saline-alkaline lands present significant challenges for agriculture, often resulting in the abandonment of valuable agricultural lands and water resources (Lin et al., 2018; Zhang et al., 2023a). Conversely, the

development of fisheries in these saline-alkaline waters provides an opportunity to transform underutilized land and water resources into productive assets, thereby promoting sustainable land use (Hu et al., 2015; Lin et al., 2018). Given this potential, it is imperative to develop and utilize this unconventional resource reasonably and effectively. Alkalinity, a critical environmental factor in saline-alkaline waters, directly influences the physiological state of aquatic animals (Qin et al., 2021). While appropriate levels of alkalinity can enhance the ability of aquatic species to withstand environmental stresses (Li et al., 2024). Usually, the pH of these alkaline salt lakes is higher than 9.0, and the

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<https://doi.org/10.1016/j.aquaculture.2025.742186>

Received 19 October 2024; Received in revised form 24 December 2024; Accepted 19 January 2025

Available online 21 January 2025

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salinity may be close to 50 % of seawater (Wilkie and Wood, 1996). This extremely alkaline environment can disrupt the osmotic balance and ion homeostasis of fish body fluids, adversely affecting their growth, antioxidant, and immune functions (Ching et al., 2009; Pang et al., 2021; Perry and Gilmour, 2006; Wongsasak et al., 2015; Yao et al., 2010). Therefore, understanding the adaptive mechanisms of fish in saline-alkaline environments, including their responses to osmotic stress and the toxic effects of high alkalinity, is crucial for their successful cultivation and survival in these challenging conditions.

The changes in environmental factors, such as salinity, alkalinity, and oxygen levels, significantly impact the normal growth and physiological metabolism of fish, leading to various forms of environmental stress. In response, fish continuously adapt by enhancing their tolerance to fluctuating water conditions in order to survive habitat changes (Zhang et al., 2024). For example, when subjected to temperature stress, heat shock proteins (HSPs) act as molecular chaperones to ensure proper protein folding, processing, and transport, thereby protecting cells from thermal damage (Aidos et al., 2020). Similarly, under hypoxic conditions, fish enhance their oxygen uptake capacity through gill remodeling to cope with oxygen deprivation (Liang et al., 2022). In contrast, saline-alkaline stress, characterized by high carbonate alkalinity and elevated pH, presents a unique challenge. This condition impedes the normal survival and reproduction of most aquatic organisms. Studies have shown that carbonate alkalinity levels of 10 to 15 mmol/L significantly inhibited the growth of Songpu mirror carp (*Cyprinus carpio Songpu*), leading to a reduction in final average body weight and weight gain rate (Fan et al., 2021). Similarly, Peng et al. (2018) reported that the survival rate of razor clams (*Sinonovacula constricta*) sharply declined when carbonate alkalinity increased from 1.22 to 45.00 mmol/L. Furthermore, crucian carp (*Carassius auratus*) exposed to 60 mmol/L NaHCO_3 over extended periods exhibited significant damage to their antioxidant systems, as well as abnormal apoptosis and immune responses in the gills (Liu et al., 2022a). Research on other species, such as Chinese Sea Bass (*Lateolabrax maculatus*) (Zhu et al., 2023) and Amur minnow (*Phoxinus lagowskii*) (Zhou et al., 2024), also indicated that alkalinity stress negatively impacts gill osmoregulation and triggers antioxidant responses. Despite these complex and variable conditions, certain species with strong environmental adaptability can survive within specific range of saline-alkaline water. Notable examples include naked carp (*Gymnocypris przewalskii*) (Tong et al., 2015), Amur Ide (*Leuciscus waleckii*) (Xu et al., 2013), and Magadi tilapia (*Alcolapia grahami*) (Wilkie and Wood, 1996), which have long inhabited extreme alkaline environments. These species adapt to saline-alkaline conditions through the regulation of physiological processes such as osmotic pressure, immune defense, and ammonia nitrogen excretion. These findings suggest that different fish species exhibit distinct stress responses and adaptation mechanisms to alkalinity stress, highlighting the necessity of further research in this area.

Gills are vital respiratory organs in aquatic animals, crucial for osmotic regulation. These functions are vital for maintaining physiological homeostasis during environmental stress and adaptive evolution (Baberschke et al., 2019; Gu et al., 2020; Wen et al., 2018). Research has shown that fish undergo stress responses in their gills when exposed to alkaline conditions, resulting in disruptions to osmotic pressure in body fluids, tissue damage, and alterations in gene expression and regulatory pathways. For example, species like *Carassius auratus gibelio* (Lu et al., 2022b) and *P. lagowskii* (Zhou et al., 2024) exhibit histopathological changes under alkalinity stress, including gill epithelial cell fusion and an increase in chloride cells. Saline-alkaline stress also affects the activity of antioxidant enzymes in fish, reducing their antioxidant capacity and inducing oxidative stress (Song et al., 2021). Studies on medaka (*Oryzias latipes*) and hybrid tilapia (*Oreochromis mossambicus* female $\times$ *O. urolepis hornorum* male) have highlighted the importance of genes related to osmotic regulation in the gills for adaptation to alkaline environments (Su et al., 2020; Yao et al., 2012). Furthermore, changes in gene expression have been observed in other freshwater fish species,

such as *G. przewalskii* (Wang et al., 2003), *Luciobarbus capito* (Shang et al., 2021) and *C. auratus* (Sun et al., 2020) under alkalinity stress. Despite these findings, Presently, research on the regulatory mechanisms and adaptation processes of marine fish under saline-alkaline stress remains limited. Understanding the response mechanisms of various cultured species, particularly marine fish, to saline-alkaline stress holds significant theoretical and practical importance.

The spotted scat (*Scatophagus argus*), a member of the order Perciformes, is widely distributed throughout the Indo-Pacific region and thrives in diverse environments, ranging from freshwater to seawater. It is particularly abundant in the southeastern coastal areas of China (Lu et al., 2022a). As a typical euryhaline fish, the spotted scat exhibits strong osmoregulatory capabilities. Previous studies have demonstrated that in response to fluctuations in environmental salinity, the osmoregulatory organs, such as the gills and kidneys, of the spotted scat undergo adaptive modifications to cope with these variations (Mu et al., 2015). Recent research utilizing RNA-Seq technology has highlighted the impact of salt stress on liver lipid metabolism in the spotted scat. Genes related to lipid metabolism have been found to be regulated in response to both low and high salinity stress, which is crucial for the fish's adaptation to varying salinity levels (Chen et al., 2023). However, there is currently a lack of studies on its adaptability under alkalinity stress conditions. Understanding the mechanisms behind its saline-alkali adaptability can offer valuable insights for the aquaculture of more euryhaline fish in saline-alkaline environments. This study delves into the effects of alkalinity stress on spotted scat at histological, biochemical, and transcriptome levels, aiming to enhance understanding of its adaptive mechanisms and survival strategies in complex water quality conditions.

2. Materials and methods

2.1. Animals and sample collection

Juvenile spotted scat were obtained from the Marine Biological Research Base of Guangdong Ocean University (Zhanjiang, China). Prior to the experiment, the fish were acclimated for two weeks in a 60 L experimental breeding tank under controlled conditions: temperature of 32 ± 1 °C, dissolved oxygen levels of 6.4 ± 0.2 mg/L, a pH range of 7.5–8.0 and ammonia nitrogen concentration below 1 mg/L. Only healthy, active, and consistent individuals were selected for the study. The experiment was approved by the Animal Care and Use Committee (IACUC) of Guangdong Ocean University and conducted in accordance with Chinese laws and biological research standards. The experimental fish were then divided into two groups: a control group and a treatment group exposed to an alkalinity of 30 mmol/L. Analytically pure NaHCO_3 (Aladdin, Shanghai) was used to achieve the desired alkalinity (Qin et al., 2021; Zhang et al., 2023b). The average body length and weight for each group were recorded as 4.46 ± 0.34 cm and 4.98 ± 0.94 g, respectively, with each group comprising three replicates. The alkalinity of the treatment group in this experiment was determined based on preliminary experimental results. Firstly, different alkalinity gradients were set for the alkalinity tolerance experiment of spotted scat (Table S1). Subsequently, the mortality of spotted scat were observed and recorded at 24, 48, and 96 h. Finally, the treatment concentration was determined based on the semi-lethal concentration (Table S2). Samples were collected at 6 time points: 0 h (C), 12 h (T12), 24 h (T24), 48 h (T48), 72 h (T72), and 96 h (T96). At each time point, five fish were randomly chosen from each replicate group for further analysis. Gill tissue samples were collected, initially fixed in Bouin's fixative, and subsequently transferred to 70 % alcohol after 24 h for paraffin sectioning and tissue observation. Additionally, gill tissue samples were cryogenically frozen in liquid nitrogen and stored at -80 °C for assessments of antioxidant capacity and gene expression.

2.2. Histological analysis of gill

Gill tissues were fixed at various time points (0 h, 12 h, 24 h, 48 h, 72 h, and 96 h). Following fixation, the samples were trimmed and dehydrated in alcohol prior to embedding in paraffin. The paraffin-embedded tissues were sectioned using an embedding machine and subsequently stained with hematoxylin-eosin. The sections were stained, then covered with neutral gum before being analyzed under an optical microscope. Specimens that met the established criteria underwent further examination using an upright microscope.

2.3. Gill biochemical parameters

The sample tissues were collected and grouped as previously described. Frozen tissue samples were thawed in a refrigerator at -20°C and then placed on ice for complete and uniform thawing. Subsequently, an appropriate volume of normal saline was subsequently added to the tissue, and the resulting supernatant was subjected to centrifugation at 3000 rpm for a duration of 10 min. All biochemical assays were performed following the guidelines provided by the manufacturer. The enzyme activities of superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GSH-Px), malondialdehyde (MDA), acid phosphatase (ACP), and alkaline phosphatase (ALP) were measured using enzyme-linked biotechnology kits (Shanghai, China).

2.4. RNA extraction, sequencing and assembly

Total RNA was extracted from the gill tissues of both the control group and the group exposed to 24 h of acute alkalinity stress using Trizol reagent (Invitrogen, Waltham, MA, USA), following the manufacturer's protocol provided by Takara (Shiga, Japan). The quality and concentration of the extracted RNA were evaluated using 1 % agarose gel electrophoresis, while RNA integrity was further verified with an Agilent 2100 Bioanalyzer (Agilent Technologies, USA). Subsequently, a cDNA library was constructed following the NEB standard protocol (Parkhomchuk et al., 2009). After construction, the library's concentration was quantified using a Qubit 2.0 Fluorometer and diluted to a final concentration of 1.5 ng/ μL . The insert size was confirmed using an Agilent 2100 bioanalyzer. Following library qualification, the library was sequenced on the Illumina NovaSeq 6000 (Illumina, USA) platform.

The measured data was processed using CASAVA base calling analysis to transform it into raw data. Any impurities in the raw data were eliminated to produce clean reads, which were then aligned to the reference genome of the spotted scat (Huang et al., 2021) using HISAT2 software (Mortazavi et al., 2008) to determine the location information of the reads. The raw data was then deposited to the National Center for Biotechnology Information (NCBI) with the accession number PRJNA1117257.

2.5. Differential expression and enrichment analysis

In this study, the FPKM method implemented in Cuffdiff software was utilized to calculate gene expression levels, and differential expression analysis was conducted using DESeq2 software (Love et al., 2014). The parameter for differentially expressed genes (DEGs) was set to $P\text{-adjust} < 0.05$ and $|\log_2\text{FoldChange}| \geq 1$ for each comparison group. Finally, Gene Ontology (GO) functional annotation and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses of DEGs were performed using clusterProfiler software version 3.8.1 (Yu et al., 2012).

2.6. Quantitative real-time PCR validation

To further validate the transcriptome, fourteen DEGs were randomly selected for qRT-PCR verification analysis. Specific primers were designed with Primer 5.0 software and synthesized by Sangon Biotech

(Shanghai) Co., Ltd., as listed in Table S3. Total RNA was extracted using the Trizol reagent, and first-strand cDNA synthesis was performed using the PrimeScript® RT Reagent Kit with gDNA Eraser (Takara, Dalian, China), in accordance with the manufacturer's instructions. qRT-PCR was conducted using the LightCycler® 480 Real-time PCR System, with β -actin serving as an internal reference to normalize gene expression levels. The qRT-PCR cycling parameters were as follows: 95°C for 5 min, 40 cycles of 95°C for 30 s and 58°C for 30 s. The relative expression of each DEG was calculated using the $2^{-\Delta\Delta\text{Ct}}$ method. Each gene and sample was analyzed in three biological and two technical replicates.

2.7. Statistical analysis

In this study, data were analyzed using SPSS 26.0. A one-way analysis of variance (ANOVA) was applied to compare differences between groups, followed by Duncan's multiple comparisons test to evaluate the significance of differences among multiple groups. A P -value of less than 0.05 was considered statistically significant. Results are presented as mean $\pm$ standard deviation ($n = 3$).

3. Result

3.1. Histological changes in gills following acute alkalinity stress

The study investigated the impact of alkalinity stress on the morphology of gill tissue in spotted scat across various exposure duration. The results showed that after 12 h of alkalinity stress, some gill lamellae on the filaments began to bend, with inconsistent bending on both sides (Fig. 1B). With 24 h of exposure, there were further inconsistent orientation of gill lamellae, increased bending compared to 12 h, and an increase in chlorine-secreting cells (Fig. 1C). By 48 h, gill lamellae bending intensified, and gill filaments narrowed compared to the control group. Continued alkalinity stress led to an increase in the diameter of chlorine-secreting cells over time, with a slight rise in their number (Fig. 1D). After 72 h and 96 h of alkalinity stress, the gill filaments continued to narrow, with some gill lamellae bending in different directions (Fig. 1E-F).

3.2. Changes in biochemical indexes of gill tissue following acute alkalinity stress

Acute alkalinity stress led to significant alterations in the biochemical indices of gill tissue in spotted scat, including the activities of SOD, CAT, MDA, GSH-Px, ACP, and ALP (Fig. 2). The activities of SOD and CAT decreased at various time points during acute alkalinity stress, with the most significant decrease observed at 96 h. The enzyme activities of SOD and CAT in the control group were consistently higher than those in the treatment groups at all time points ($P < 0.05$) (Fig. 2A-B). Conversely, the activities of GSH-Px, ACP, ALP, and MDA exhibited a progressive increase with prolonged exposure to alkalinity stress, attaining significantly higher levels compared to the control group ($P < 0.05$) (Fig. 2C-F).

3.3. Transcriptome sequencing analysis

3.3.1. Sequencing data filtering and quality control

Six transcriptome libraries were generated using gill tissue samples from spotted scat exposed to 24 h of acute alkalinity stress, yielding a total of 271,470,014,898 raw reads. After quality control procedures, 261,196,516 clean reads were obtained, averaging 43.53 million clean reads per sample. The raw data consisted of over 6.1 billion bases, which were refined to over 5.9 billion bases after filtering. Quality assessments indicated Q20 values exceeding 97 %, Q30 values surpassing 92 %, while GC content ranged from 48.31 % to 49.86 % (Table 1). These clean reads were aligned to the spotted scat genome sequence, achieving an

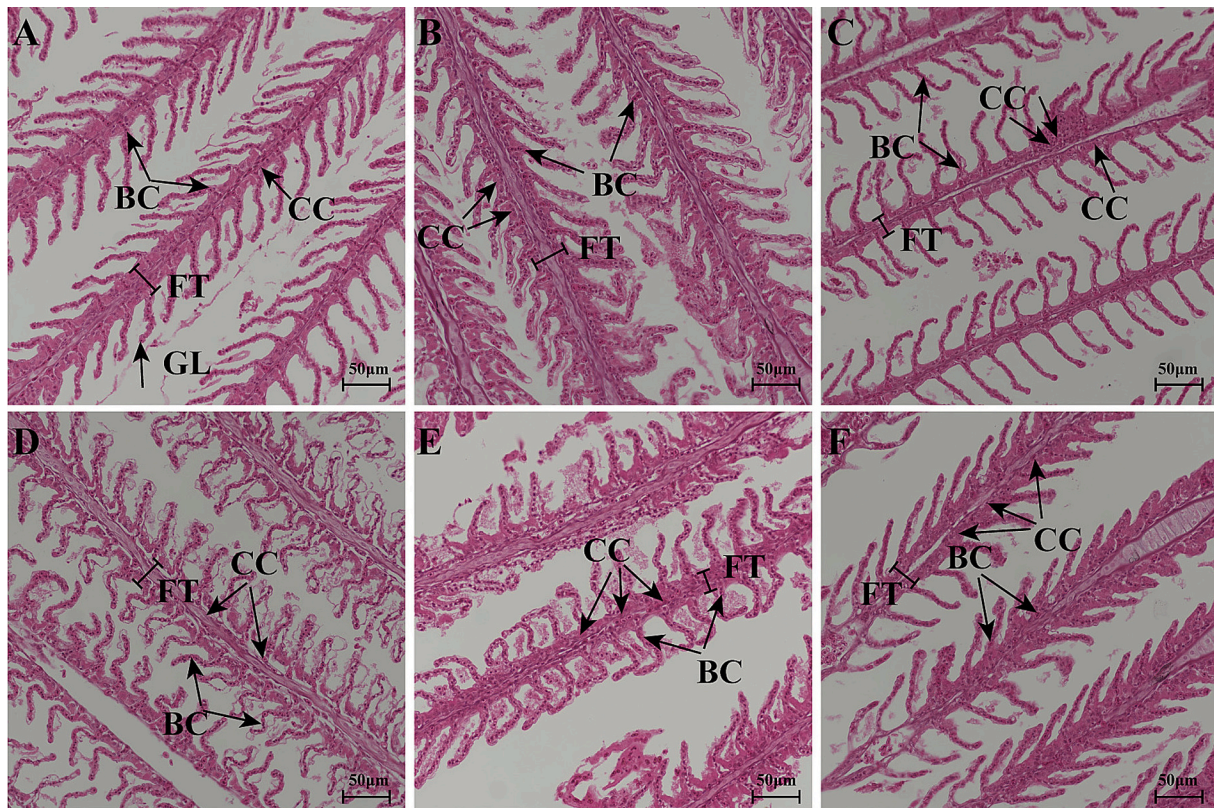


Fig. 1. Changes in gill tissue structure of spotted scat under acute alkali stress at various time points: (A) control group, (B) 12 h treatment, (C) 24 h treatment, (D) 48 h treatment, (E) 72 h treatment, and (F) 96 h treatment. BC: blood cells; CC: chloride-secreting cells; GL: gill lamellae; FT: gill filament thickness.

alignment ratio of more than 89.94 % (Table S4). The high sequencing quality of these data ensures the reliability of the subsequent transcriptome analysis.

3.3.2. Differential expression analysis

Under acute alkalinity stress, a total of 513 DEGs were identified when comparing the 24 h alkalinity stress group to the control group. Of these, 292 genes were found to be up-regulated, while 221 genes were down-regulated (Fig. 3A). The top 10 up-regulated and downregulated genes from both the control and treatment groups are presented in Table 2. Furthermore, hierarchical clustering analysis was conducted on the DEGs to visualize expression differences between the two conditions. The findings clearly indicated that alkalinity stress significantly influenced gene transcription within the gill tissue of spotted scat, resulting in substantial variations in expression levels between the two groups (Fig. 3B).

3.3.3. Pathway enrichment of DEGs

To investigate the biological significance of DEGs, we conducted GO enrichment analysis. The results showed that DEGs were significantly enriched in categories such as cellular component organization or biogenesis, intracellular signal transduction, and proteolysis (Fig. 4). Within the cellular component category, subcategories included cytoplasm, organelle parts, and nucleus. In the molecular function category, the majority of DEGs were associated with catalytic activity and binding (Fig. 4A). Furthermore, the circular diagram of GO enrichment analysis highlighted a substantial enrichment of up-regulated genes across multiple GO terms. These DEGs play key roles in ion transport, organelle function, enzyme catalysis, and binding, significantly contributing to environmental adaptation (Fig. 4B). To further investigate the key genes and pathways in the gills of spotted scat under alkalinity stress, we functionally enriched the DEGs. Under acute alkalinity stress, DEGs were significantly enriched in pathways related to drug metabolism-

other enzymes and drug metabolism-cytochrome P450, as well as arachidonic acid and arginine and proline metabolism (Fig. 4C).

3.3.4. Immune metabolic pathways

To further investigate the impact of spotted scat response to alkalinity stress, we analyzed the expression levels of key DEGs involved in various metabolic pathways, including drug metabolism-other enzymes, drug metabolism-cytochrome P450, arachidonic acid, and arginine and proline metabolism (Fig. 5A-B). Most genes associated with immune metabolism exhibited significant up-regulation following acute alkaline stress treatment, including *gstt3*, *aldh4a1*, *gsta4*, *ugt2a1*, *ptgr1*, *dpydb*, *mao*, *xdh*, *gsta3*, *prxl2b*, *amd1*, *ugt1a2*, *zgc*, and *pla2g1*. In contrast, genes such as *rrm2*, *pycr2*, *fmo5*, *arg2-a*, and *ptgs2* were significantly down-regulated. These findings further underscore the impact of alkalinity stress on gene expression levels in the gills of spotted scat.

3.4. Validation of DGEs with qRT-PCR

To verify the transcriptome sequencing results, 14 DEGs were randomly selected from gill tissues for qRT-PCR confirmation, including 10 up-regulated and 4 down-regulated genes (Fig. 6). The qRT-PCR results corroborated the RNA-seq data, thereby confirming the reliability and accuracy of the findings from the transcriptome sequencing analysis.

4. Discussion

Saline-alkaline waters present a challenging environment that impedes the survival of most aquatic animals, leading to prolonged barren conditions that significantly restrict the development of the fishery economy (Sharma and Minhas, 2005). The elevated levels of salinity and alkalinity pose substantial threats to fish survival, necessitating an investigation into the molecular mechanisms that allow aquatic animals

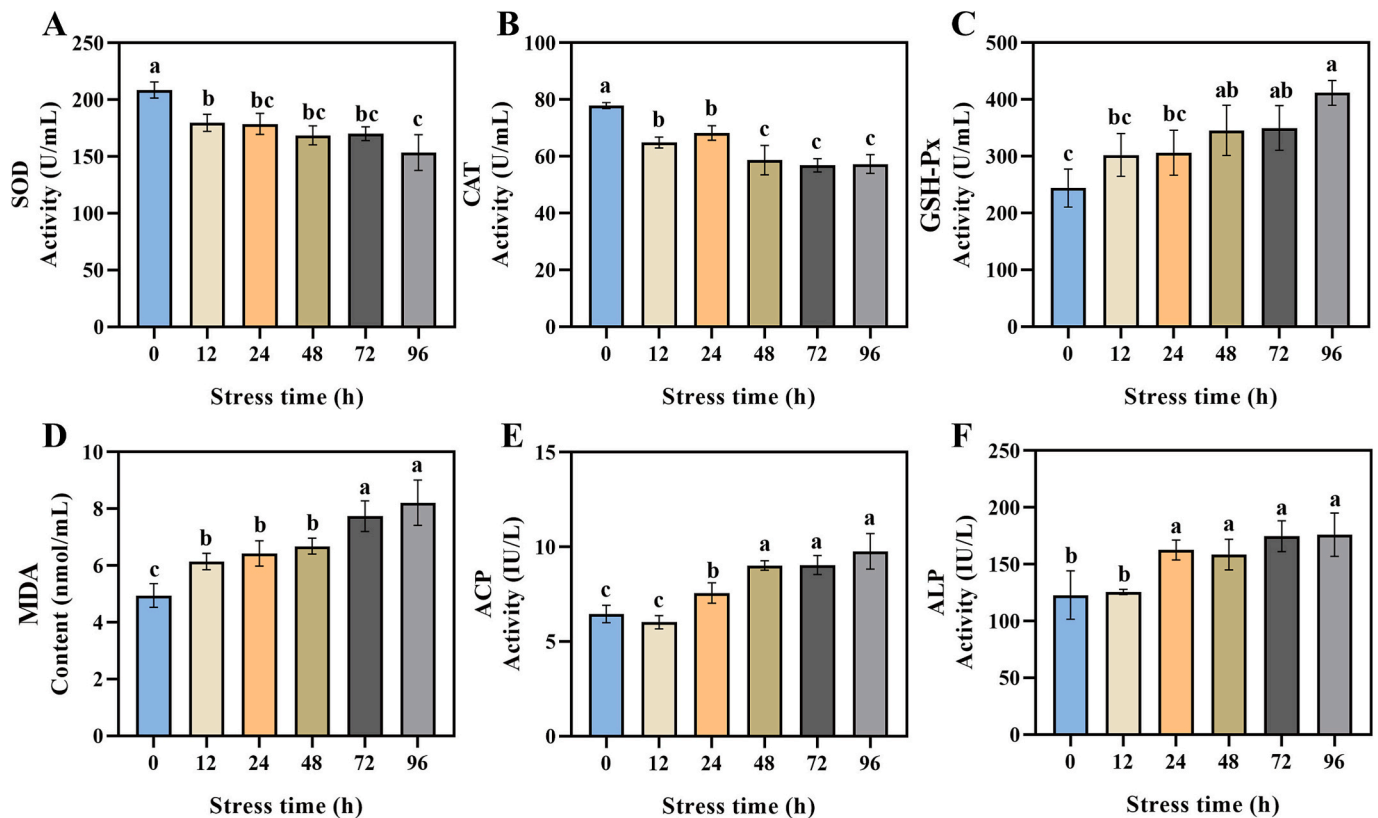


Fig. 2. Effects of alkali stress on biochemical indexes of gill tissue in spotted scat over various time intervals (0 h as control, 12 h, 24 h, 48 h, 72 h, and 96 h as treatment groups). Data are presented as mean $\pm$ SD ($n = 3$), with vertical line representing the standard error. Statistical significance is denoted by different letters, indicating a significance level of $P < 0.05$.

Table 1

Statistics of transcriptome sequencing data of gill tissue of spotted scat.¹

Sample	Raw reads	Raw bases (G)	Clean reads	Clean bases(G)	Q20 (%)	Q30 (%)	GC pct
C1gi	41,239,478	6.19	39,485,786	5.92	97.28	92.82	48.49
C2gi	46,589,486	6.99	44,910,906	6.74	97.46	93.15	49.11
C3gi	42,393,742	6.36	40,859,426	6.13	97.31	92.84	48.34
T1gi	41,981,904	6.30	40,189,054	6.03	97.35	92.96	48.61
T2gi	49,871,384	7.40	48,814,746	7.32	97.65	93.57	49.86
T3gi	49,394,020	7.41	46,936,598	7.04	97.44	93.11	48.31

¹ Note: Three replicates were selected from the gill tissue of the control group, labeled as Control 1–3 (C1, C2, C3), and three replicates were selected from the 24-h alkali stress treatment group, labeled as Treatment1–3 (T1, T2, T3).

to adapt to these conditions. This study focuses on the euryhaline marine fish, spotted scat, with the aim of elucidating its adaptive mechanisms in response to alkalinity stress through histological, biochemical, and transcriptomic analyses. Understanding these mechanisms is essential for advancing aquaculture in saline-alkaline environments.

4.1. Acute alkalinity stress treatment induces gill histological damage

Fish exhibit a high sensitivity to environmental stressors, and numerous studies have demonstrated that stress can rapidly induce significant damage in gill tissues, with alterations often observable within hours or days (LeBlanc et al., 2010; Nilsson et al., 2012). Histopathology can directly and accurately reveal the pathological changes of tissues and cells, and is a reliable indicator to determine the impact of pollution on aquatic organisms (Liu et al., 2022b). In this study, we observed cell contraction in the gill filaments, bending and disorientation of the gill lamellae, and swelling at the base of the filaments after 12, 24 and 48 h of alkalinity stress. These changes became increasingly pronounced over time when compared to the control group. This

indicates that acute alkalinity stress treatment resulted in notable histological damage in the gills of the stressed group. After 72 and 96 h of alkalinity stress, the bending and misorientation of gill lamellae were reduced. Additionally, a significant accumulation of eosinophils with abundant cytoplasm was also detected in the gill tissue, suggesting that these cells are rich in carbonic anhydrase and sodium/potassium transport ATPase, are involved in the transport of CO_3^{2-} , K^+ , and Na^+ across the cell membrane to maintain osmotic pressure stability (Hwang et al., 2011; Evans et al., 2005). We hypothesize that under alkalinity stress, gill tissue damage and elevated ion transport pressure lead to enhanced chloride cell function, thereby increasing the fish's capacity for osmotic adjustment. Similar histological changes and adaptations have been reported in other fish species, such as *C. gibelio* (Lu et al., 2022b), *P. lagowskii* (Zhou et al., 2024), *L. maculatus* (Zhang et al., 2023a), and grass carp (*Ctenopharyngodon idella*) (Wen et al., 2023). The findings indicate that alkalinity stress can cause damage to the gill tissue of spotted scat and underscore the importance of adaptive mechanisms, such as the increase in chloride-secreting cells, in facilitating fish survival in harsh environments.

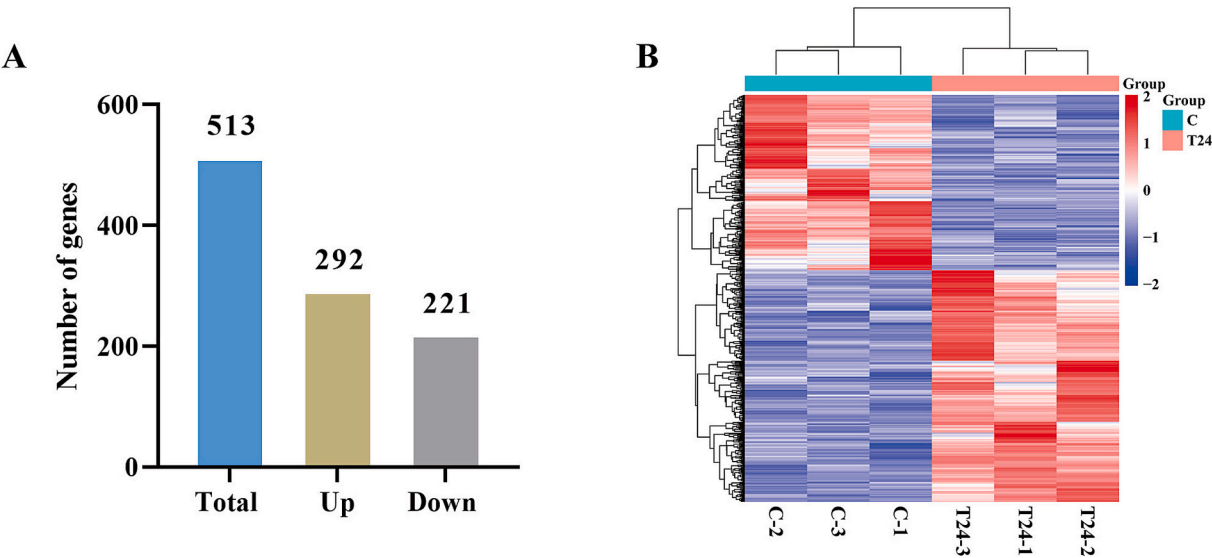


Fig. 3. (A) The number of DEGs between the control (C) group and the 24-h treatment (T24) group. The vertical axis represents the number of genes, with yellow indicating up-regulated DEGs and gray indicating down-regulated DEGs. (B) Hierarchical clustering heat map analysis of all DEGs in the C group and the T24 group. The x-axis represents the different treatment times of the C group and the T24 group, while the y-axis represents each DEG. The $\log_2$ (FPKM + 1) values were normalized, transformed (scaled), and clustered. Red indicates high gene expression, and blue indicates low gene expression. The color gradient from red to blue indicates that $\log_2$ (FPKM + 1) values range from large to small. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Top 20 DEGs in control group vs. treatment group of spotted scat under acute alkali stress.

Gene ID	Gene Name	Gene Annotation	$\log_2$ (FC)
EVM0000849	<i>nlrc3</i>	NLR family CARD domain-containing protein 3-like	6.95
EVM0022194	<i>cry1</i>	Cryptochrome-1	4.57
EVM0004739	<i>nr1d1</i>	Nuclear receptor subfamily 1 group D	4.44
EVM0008364	<i>aadac</i>	Arylacetamide deacetylase	4.25
EVM0015572	<i>slc14a2</i>	solute carrier family 14 member 2	4.08
EVM0012409	<i>dbpa</i>	D site albumin promoter binding	3.77
EVM0006734	<i>cry1b</i>	cryptochrome circadian regulator 1b	3.44
EVM0020556	<i>rims4</i>	regulating synaptic membrane exocytosis 4	3.18
EVM0020556	<i>rab11fip5</i>	RAB11 family interacting protein 5	3.11
EVM0000340	<i>ddb2</i>	DNA damage-binding protein 2	3.05
EVM0014461	<i>mid1ip1b</i>	Mid1-interacting protein 1-B	-4.77
EVM0009045	<i>clocka</i>	clock circadian regulator a	-4.09
EVM0017874	<i>cpo</i>	Carboxypeptidase O	-3.97
EVM0013478	<i>nfil3</i>	Nuclear factor interleukin-3-regulated protein	-3.73
EVM0009074	<i>cc120b</i>	chemokine (C–C motif) ligand 20b	-3.63
EVM0008204	<i>fkbp5</i>	FKBP prolyl isomerase 5	-3.63
EVM0000528	<i>rag1</i>	recombination activating 1	-3.54
EVM0005323	<i>c1ql4</i>	Complement C1q-like protein 4	-3.50
EVM0017011	<i>illr4</i>	immune-related, lectin-like receptor 4	-3.32
EVM0006733	<i>deptor</i>	DEP domain-containing mTOR-interacting protein	-2.77

4.2. Oxidative damage of gill tissue caused by acute alkalinity stress

Environmental factors are well-known to induce antioxidant defense mechanisms in organisms. However, under extreme conditions, the delicate balance of these antioxidant systems can be disrupted, leading to excessive production of reactive oxygen species (ROS). This overproduction of ROS can result in oxidative damage to lipids, proteins, DNA, and carbohydrates (Singh et al., 2021). SOD and CAT are key enzymes that constitute the first line of defense against ROS, playing a crucial role in protecting organisms from oxidative stress (Žikić et al., 2001). In this study, the activities of SOD and CAT decreased at various

time points during acute alkalinity stress, with the most significant decrease observed at 96 h. The results showed that with the prolongation of stress time, the gill tissue was seriously damaged by oxidation. This reduction in antioxidant enzyme levels aligns with findings from previous studies on *C. auratus* (Liu et al., 2022a) and *L. capito* (Shang et al., 2021). The diminished activity of SOD and CAT likely leads to an accumulation of free radicals and peroxides, which can trigger oxidative stress and cause further cellular damage (Baysoy et al., 2012). GSH-Px is another essential enzyme in the antioxidant defense system, responsible for removing peroxides by utilizing GSH as a substrate to degrade hydrogen peroxide (H₂O₂) and organic peroxides. This function is crucial for providing protection against ROS and oxidative stress (Cheff et al., 2018). When the activities of SOD and CAT are compromised under alkaline stress, an increase in GSH-Px activity can compensate by clearing the accumulated H₂O₂ and organic peroxides, thereby safeguarding tissues from oxidative damage (Fan et al., 2021). In this study, GSH-Px activity gradually increased with the prolongation of alkalinity stress time, reaching its peak at 96 h. This increase suggests that alkali stress may enhance the body's metabolic regulation, particularly in response to oxidative stress challenges. The elevated GSH-Px activity likely serves as a protective mechanism against oxidative damage induced by hydrogen peroxide and organic peroxides, consistent with previous studies on the effects of saline-alkaline stress on *C. auratus* (Ding et al., 2023).

MDA is a significant byproduct of lipid peroxidation and serves as a key indicator for assessing ROS levels and oxidative stress in organisms (Paust et al., 2011). In this study, the concentration of MDA gradually increased with the duration of alkalinity stress, significantly surpassing the control group, especially at 96 h. This increase is likely due to the excessive ROS production triggered by alkali stress, resulting in substantial lipid peroxidation and subsequent oxidative stress within the gill tissues, with the most intense peroxidation occurring at 96 h. These findings align with previous research on *C. gibelio*, which also reported a marked rise in MDA levels following saline-alkaline stress, resulting in ROS accumulation and oxidative damage (Lu et al., 2022b). Furthermore, the phosphorylation and dephosphorylation of ACP and ALP are critical processes in the body's detoxification mechanisms and serve as important indicators of immune function (Dash et al., 2014). In this

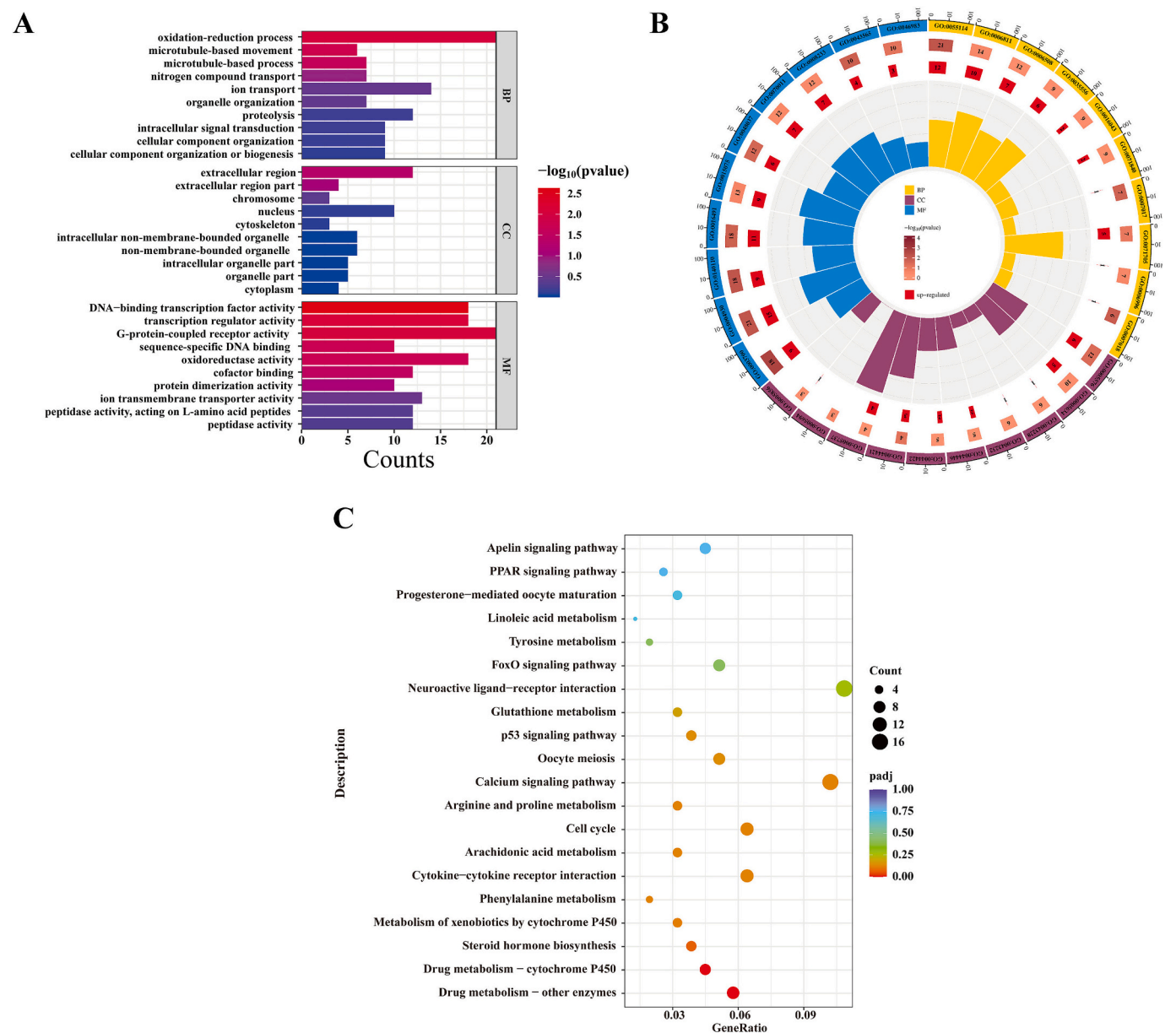


Fig. 4. (A) GO enrichment analysis of DEGs between the control (C) group and the 24-h treatment (T24) group. (B) GO enrichment analysis circle diagram of DEGs in the C group and the T24 group. Yellow represents biological processes (BP), dark purple represents cell component (CC), blue represents molecular function (MF), and red represents up-regulated gene. (C) KEGG enrichment analysis of DEGs in the C group and the T24 group, with the dot color closer to red indicating a more significant difference. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

study, at the early stage of alkalinity stress (12h), the activities of ACP and AKP were not significantly different from those of the control group. we observed a significant increase in the activities of ACP and ALP with prolonged exposure to alkalinity stress and remained higher than those of the control group, peaking at 96 h. This finding aligns with observations made in *P. lagowskii* (Zhou et al., 2024) and *Labeo rohita* (Patel et al., 2022), suggesting that fish may adapt to environmental changes through a dephosphorylation-mediated stress response. In conclusion, the observed decline in SOD and CAT activities, coupled with the increase in MDA concentrations, indicates that spotted scat experience significant oxidative damage under acute alkalinity stress. Conversely, the elevated activities of GSH-Px, ACP, and ALP suggest the presence of a compensatory mechanism that may mitigate the adverse effects of a hypertonic environment. These adaptive responses underscore the complex physiological strategies that fish employ to survive and maintain homeostasis under stressful conditions, thereby providing valuable

insights into their resilience in fluctuating environments.

4.3. Alkalinity treatment can regulate the molecular mechanism of the body's response to alkalinity stress

Recent advancements in transcriptome technology have significantly improved our understanding of fish responses to environmental stress, elucidating comprehensive gene expression patterns and associated physiological responses under specific conditions (Smith et al., 2013). In this study, transcriptome analysis was conducted to explore the key molecular mechanisms in the gill tissue of spotted scat after 24 h of acute alkalinity stress. The analysis identified a substantial number of DEGs, indicating considerable transcriptional regulation triggered by acute alkali stress. Specifically, a total of 513 DEGs were detected, comprising 292 up-regulated genes and 221 down-regulated genes. Among these DEGs, a notable number of these DEGs were associated with immune

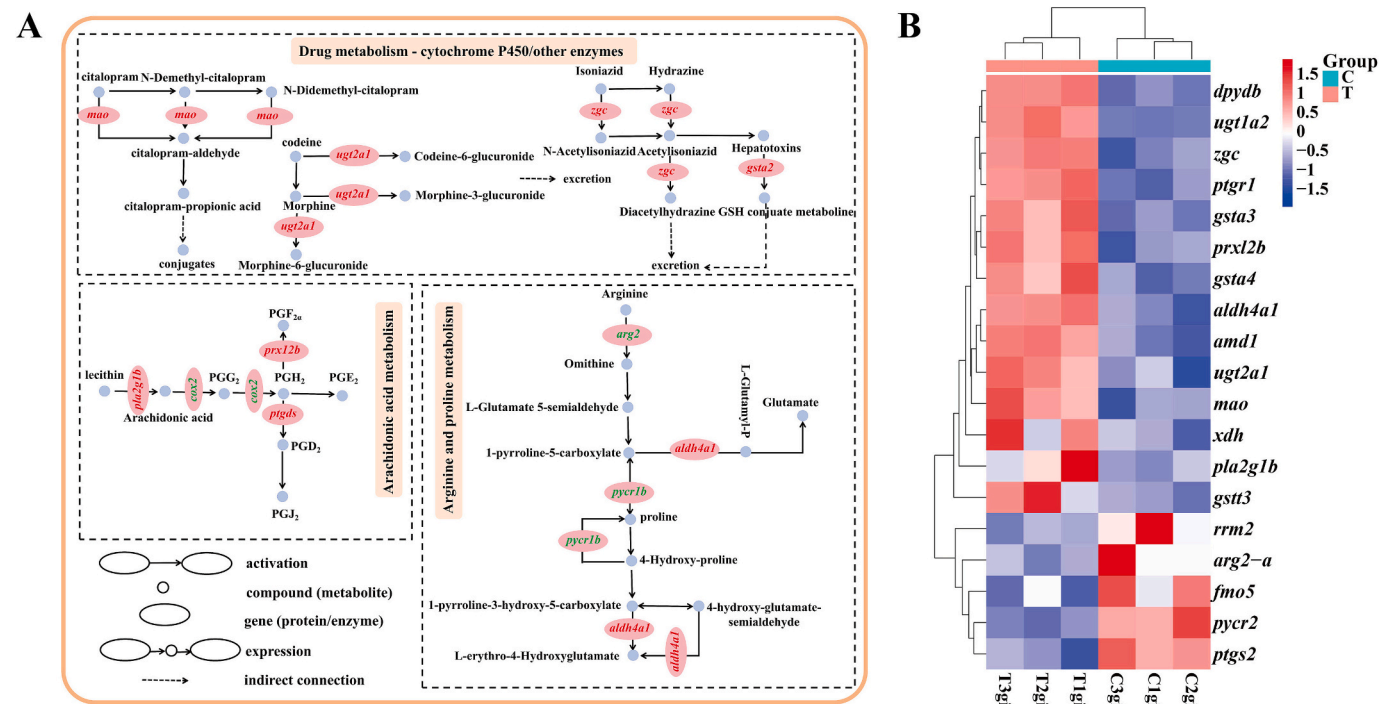


Fig. 5. (A) Key pathways and genes involved in the response to alkali stress. Red font indicates upregulated genes, while green font indicates downregulated genes. (B) Hierarchical clustering analysis of all differential genes in KEGG enrichment pathway. The x-axis represents the sample names, and the y-axis represents the gene names. The $\log_2$ (FPKM +1) values were normalized and transformed (scaled) and clustered. The redder the color, the higher the expression level, while bluer colors indicate lower expression levels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

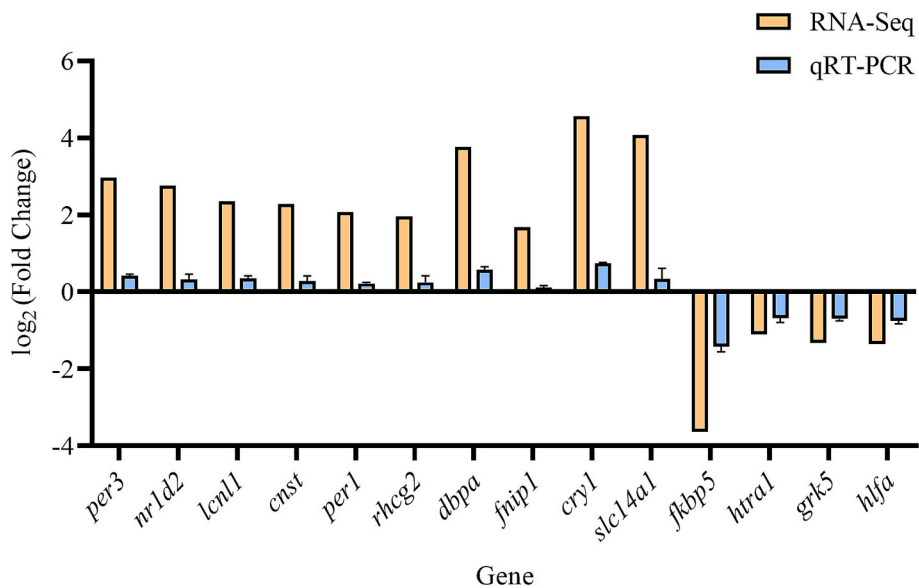


Fig. 6. Comparison of gene expression levels in the gills of spotted scat as determined by RNA-seq and qRT-PCR. The data are expressed as mean $\pm$ SD of three replicates (n = 3).

functions. Further KEGG enrichment analysis revealed that the DEGs related to the stress response were involved in pathways related to drug metabolism, particularly those concerning cytochrome P450 and other drug metabolism enzymes. Additionally, the metabolism processes of arachidonic acid, arginine, and proline were emphasized as crucial for adaptation to alkalinity stress.

Carbonate alkalinity is known to induce respiratory alkalosis in fish (Wilkie et al., 1994). To resist environmental stress and maintain homeostasis, organisms often rely on detoxification enzymes, which play a

crucial role in biotransformation processes (Šrejber et al., 2018). Among these enzymes, UDP-glucuronosyltransferases (Ugts) are essential for detoxifying xenobiotic substances, converting lipophilic small molecules into water-soluble metabolites that can be excreted from the body (Christen and Fent, 2014; Sneitz et al., 2013). In this study, we observed significant enrichment in pathways related to drug metabolism, particularly the cytochrome P450 system, along with other drug metabolism enzymes. There was notable up-regulation of UDP glucuronosyltransferase (UGT) family genes, such as *ugt1a2* and *ugt2a1*, as well as

glutathione transferase (GST) family genes, including *gsta3*, *gsta4*, and *gstt3*. UGT-catalyzed glucuronidation is a vital metabolic pathway for maintaining homeostasis and plays a critical role in the detoxification and excretion processes in fish (Udomuksorn et al., 2007). Previous studies have shown that in conditions of stress, such as starvation in mice, UGT gene expression in the liver is up-regulated to regulate hormone levels and eliminate harmful substances (Xu et al., 2012). Conversely, in spotted scat under low salinity conditions, genes like *ugt2a1* and *ugt2a2* were found to be down-regulated, which likely represents an adaptive response to reduce the metabolism of small lipophilic molecules (Chen et al., 2023). In this study, the up-regulation of *ugt1a2* and *ugt2a1* genes suggests that fish protect themselves from the accumulation of foreign substances through enhanced glucuronidation processes (Wang et al., 2014). Glutathione transferases (GSTs) also play a dual role under abiotic stress conditions. They function as antioxidant enzymes and catalyze the conjugation of glutathione (GSH) to both endogenous and exogenous compounds, facilitating detoxification (Booth et al., 1961; Yang, 2002). For example, GST genes in *Spodoptera litura* were significantly induced and up-regulated following exposure to tomatine, suggesting that GST family genes are crucial for defending against TOM (Li et al., 2019). In the current study, significant expression of the *gsta3*, *gsta4*, and *gstt3* genes in the gill tissues of spotted scat suggests that the gill tissue, an organ crucial for osmoregulation, also plays an important role in detoxification under alkalinity stress. GSTs likely contribute to the organism's adaptation by catalyzing the binding of GSH to electrophilic compounds, thereby neutralizing toxic substances and protecting cells from oxidative damage. Overall, these findings indicate that alkali stress triggers the activation of drug metabolism pathways, leading to enhanced expression of UGT and GST subtypes in the gill tissue of spotted scat. This up-regulation is likely a critical adaptive response that helps the fish cope with the toxic effects of the stressful environment. It is speculated that overexpression of these detoxification-related genes may enhance the tolerance of spotted scat to alkalinity stress, highlighting their positive regulatory role in the fish's stress response mechanisms.

The arachidonic acid (AA) pathway is crucial in various physiological processes, particularly in mediating inflammation (Badimon et al., 2021). AA is metabolized by cyclooxygenase (COX)-1/2 and specific prostaglandin synthase enzymes into prostaglandins such as PGE₂ and PGD₂, which play key roles in regulating lipid metabolism, biological processes, and immune responses (Heneka, 2015). In this study, we observed a significant down-regulation of the *cox-2* gene and an up-regulation of the *ptgds* gene in the gill tissues of spotted scat after 24 h of acute alkalinity stress treatment. *Cox-2* is a key enzyme that catalyzes the conversion of AA into various prostaglandins (PGs) (Fitzpatrick, 2004). Under stress conditions, *cox-2* is typically up-regulated, leading to the production of PGs that induce inflammation and tissue damage (Simmons et al., 2004). For example, Liu et al. (2022a) reported that *cox-2* expression was significantly higher in the gill tissues of *C. auratus* subjected to carbonate alkalinity stress, resulting in accelerated conversion of AA and increased inflammation. However, in this study, the down-regulation of *cox-2* suggests that acute alkalinity stress may have caused dysfunction in the gill tissues of spotted scat, leading to a reduction in the inflammatory response by inhibiting *cox-2* expression. Prostaglandins, which are metabolites of arachidonic acid, are produced and released in response to various physiological and pathological stimuli and are critical for maintaining local homeostasis in the body (Narumiya, 2007). *ptgds* is an important regulatory molecule in the inflammatory response (Xue et al., 2014). During inflammation, immune cells can convert PGH₂ into PGD₂, leading to the up-regulation of *ptgds* and contributing to the inflammatory process. In this study, the up-regulation of *ptgds* indicates that acute alkalinity stress stimulated the expression of *cox-2* and promoted the production of PGs, despite the overall down-regulation of *cox-2*. In summary, acute alkalinity stress appears to cause tissue damage and induce an inflammatory response in spotted scat, particularly in the gill tissues directly exposed to alkaline

water. The changes in gene expression suggest that while the inflammatory response is initially activated, there may also be an enhancement of autoimmunity through increased activity of anti-inflammatory factors, allowing the organism to better cope with changes in the external environment. These findings provide valuable insights into the complex interplay between inflammatory and anti-inflammatory responses in fish under environmental stress.

5. Conclusion

This study comprehensively examined the effects of alkalinity stress on the tissue structure, antioxidant capacity, and gene expression of spotted scat gills through histological, biochemical, and transcriptomic analyses. The findings demonstrate that alkalinity stress induces oxidative stress, leading to excessive ROS production and disrupting the balance within antioxidant systems, which in turn causes structural and functional damage to the gill tissue. Additionally, acute alkalinity stress alters the gene and enzyme activities associated with immune and metabolic pathways in the gill tissue of spotted scat. This research provides a molecular perspective on the tolerance of spotted scat to alkalinity stress and offers valuable insights for future studies on the mechanisms underlying alkali resistance in this species.

Ethics approval

The Guangdong Ocean University Animal Research and Ethics Committee authorized each and every experimental procedure used in this work (NIH Pub. No.85–23, amended 1996). There are no protected or endangered species included in this study. Every attempt was made to reduce the animals' suffering. Every experiment was carried out in compliance with all applicable laws and rules.

CRediT authorship contribution statement

Lei Yang: Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – original draft. **Adili Abudu:** Supervision, Investigation, Funding acquisition. **Kecheng Zhu:** Investigation, Data curation. **Tong Han:** Investigation, Data curation. **Cunyu Duan:** Investigation, Data curation. **Yu Chen:** Investigation, Data curation. **Xiaolong Li:** Investigation, Data curation. **Gang Shi:** Data curation. **Chunhua Zhu:** Funding acquisition. **Changxu Tian:** Supervision, Funding acquisition, Writing – review & editing. **Guangli Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw data was then deposited to the National Center for Biotechnology Information (NCBI) with the accession number PRJNA1117257.

Acknowledgments

This research was supported by grants from the National Natural Science Foundation of China (32473155, 32201420 and 32002367), Key Research and Development Task Special Project of the Xinjiang Uygur Autonomous Region (2024B02014), the Development Fund of Key Laboratory of Fisheries Resources Exploitation and Utilization in South China Sea, Ministry of Agriculture and Rural Affairs (FREU2024-05), the Guangdong Provincial Science and Technology Program (2023B0202010005), Guangdong Basic and Applied Basic Research

Foundation (2023A1515012880 and 2024A1515010328), the Department of Education of Guangdong Province (2023KTSCX042), the Zhanjiang Science and Technology Plan Project (2022A01046).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2025.742186>.

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